

KUOPIO RITONIEMI, Finland

WMO: 027320

Lat: 62.799N

Lon: 27.905E

Elev: 285

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-11.5	-23.4	1.5	-17.6	-16.5	2.3	-11.3	24.5	29.4	23.0	28.4	2.7	210	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	79.8	65.9	76.1	64.0	73.1	62.4	68.1	75.1	66.4	73.7	64.4	71.0	7.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.7	71.6	63.5	88.7	70.0	61.3	81.9	68.3	32.5	74.4	31.1	74.0	29.6	70.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.5	18.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	14.4	18.8	25.0	36.7	49.9	58.1	63.9	61.7	52.6	39.9	31.9	22.6
	DBStd	18.94	14.18	13.51	9.73	5.80	8.68	6.85	6.03	5.68	5.68	6.23	7.41	11.93
	HDD50	5024	1105	874	776	398	111	9	0	1	40	318	542	850
	HDD65	9338	1570	1294	1241	848	471	227	94	133	374	779	992	1315
	CDD50	1275	0	0	0	0	107	252	431	364	117	4	0	0
	CDD65	113	0	0	0	0	3	19	60	31	0	0	0	0
	CDH74	544	0	0	0	0	25	89	300	130	0	0	0	0
	CDH80	85	0	0	0	0	1	11	47	26	0	0	0	0
Wind	WSAvg	9.2	8.8	8.9	9.7	8.3	8.1	9.2	7.9	8.9	9.6	10.7	10.4	9.7
Precipitation	PrecAvg	24.8	1.7	1.4	1.4	1.5	1.7	2.5	3.1	3.0	2.4	2.1	2.2	1.9
	PrecMax	29.7	3.0	2.8	3.0	2.4	4.9	4.7	5.7	4.5	4.5	3.9	4.6	3.3
	PrecMin	18.0	0.5	0.2	0.5	0.7	0.6	0.8	0.5	0.7	1.0	0.9	0.4	0.7
	PrecStd	3.1	0.7	0.7	0.6	0.5	1.1	1.0	1.2	1.0	1.0	0.9	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	40.3	46.0	56.7	77.5	81.6	84.5	83.4	69.2	55.5	48.4	42.4
		MCWB	34.9	36.8	38.5	46.1	59.2	65.9	67.2	68.5	60.1	51.2	45.9	40.8
	2%	DB	35.0	37.6	41.5	52.5	73.0	76.4	80.9	77.2	64.6	52.8	45.4	39.1
		MCWB	33.9	35.2	36.0	43.6	57.3	64.0	66.9	65.6	58.3	50.3	43.6	37.8
	5%	DB	33.5	35.8	38.7	48.8	69.1	72.6	77.3	73.4	62.0	50.8	42.7	36.4
		MCWB	32.4	34.3	34.4	41.6	56.8	61.7	65.3	63.2	57.2	48.7	41.8	35.0
	10%	DB	32.2	33.9	36.6	45.7	63.9	69.3	73.9	70.3	60.1	48.6	40.4	34.5
		MCWB	31.5	33.0	33.7	39.3	53.4	59.8	63.9	62.2	56.2	46.6	39.4	33.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.6	37.3	39.1	48.1	62.1	68.6	70.6	71.0	61.1	52.7	46.8	41.1
		MCDB	36.0	39.4	43.6	52.1	73.0	76.3	78.9	81.6	65.0	54.5	48.5	42.7
	2%	WB	34.1	35.7	36.8	44.6	59.4	66.2	68.6	66.7	59.8	50.9	43.7	37.7
		MCDB	34.7	37.3	40.1	52.5	68.7	74.9	75.9	73.7	63.0	52.4	45.2	38.8
	5%	WB	32.9	34.3	35.3	42.0	57.3	62.9	67.2	64.8	58.3	49.0	41.8	35.1
		MCDB	33.7	35.4	37.8	47.1	67.4	70.7	74.4	71.2	61.4	50.6	42.7	36.1
	10%	WB	31.7	33.1	34.1	40.1	54.5	60.3	65.6	63.2	56.7	46.6	39.6	33.6
		MCDB	32.5	34.0	36.1	44.4	63.5	67.9	72.7	69.1	59.7	48.4	40.2	34.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.9	14.0	13.2	14.3	12.1	11.7	10.4	8.1	6.2	5.0	7.3
		MCDBR	6.3	8.5	13.3	19.0	21.1	16.7	16.9	14.9	11.1	7.8	6.8	6.3
		MCWBR	6.1	7.4	9.4	11.7	9.6	7.3	6.4	6.0	6.4	6.0	6.6	5.9
	5% WB	MCDBR	6.0	7.8	10.8	16.1	20.6	15.5	15.0	13.0	10.1	7.7	6.8	6.1
		MCWBR	6.0	7.0	8.3	10.5	9.8	7.3	7.0	6.0	6.5	6.6	6.7	5.8
Clear-Sky Solar Irradiance	taub	0.231	0.287	0.313	0.347	0.345	0.343	0.367	0.351	0.328	0.313	0.249	N/A	
	taud	2.009	2.115	2.186	2.347	2.468	2.499	2.447	2.478	2.531	2.400	2.100	N/A	
	Ebn at Noon	128	203	247	266	277	279	268	263	248	200	120	N/A	
	Edh at Noon	10	21	30	31	30	30	31	27	22	16	10	N/A	
All-Sky Solar Radiation	RadAvg	61	240	725	1161	1524	1673	1594	1219	681	264	67	27	
	RadStd	9	57	90	138	168	102	157	140	77	45	12	3	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	-338	N/A	N/A	N/A

Nomenclature: See separate page